BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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> **Due Date**
>
> Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Documents/GitHub/hw3-sarah-nathan-1`
```

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day-1, and the decay rates of CBOD and NBOD are are 0.35 and 0.25 day-1, respectively. The river's saturated dissolved oxygen concentration is 10m g/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance `d` km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m3/d	10,000 m3/d	15,000 m3/d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

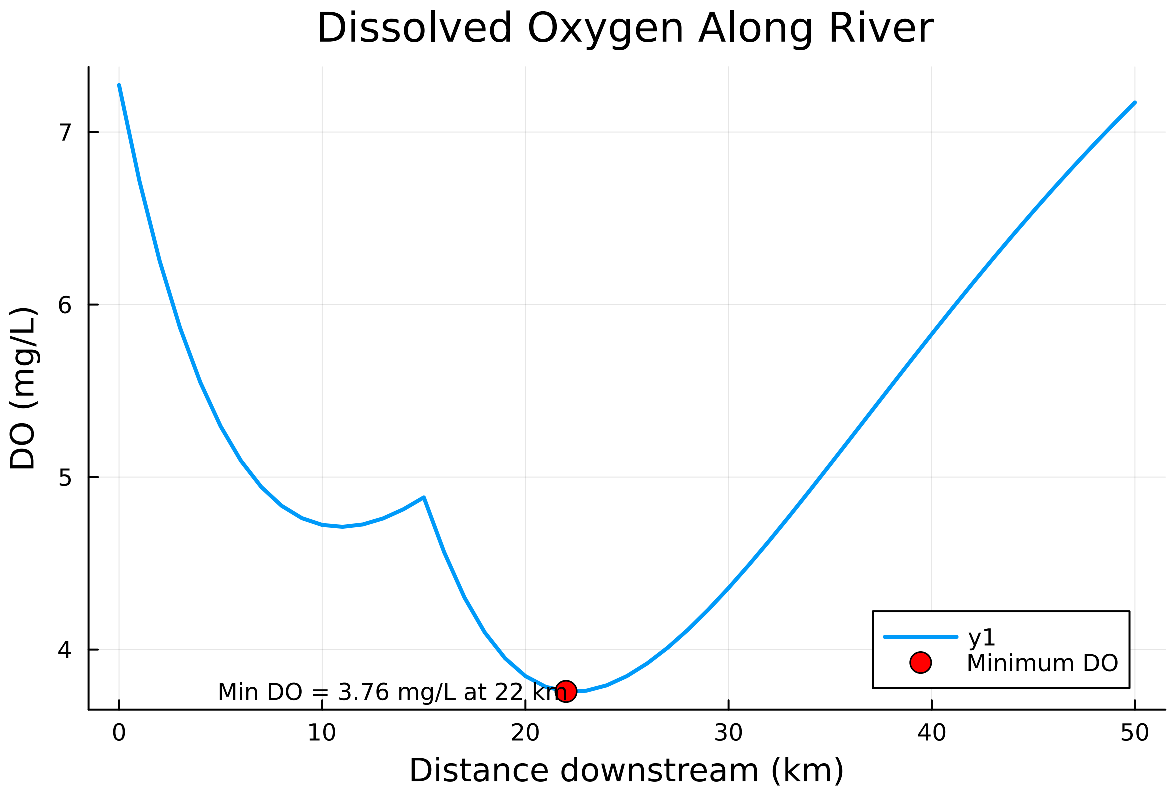
Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
# Parameters
v = 6            # River velocity (km/d)
Cs = 10          # Saturated DO concentration (mg/L)

ka = 0.55        # Reaeration rate (d^-1)
kc = 0.35        # CBOD decay rate (d^-1)
kn = 0.25        # NBOD decay rate (d^-1)

Qr = 100000      # River inflow (m^3/d)
Q1 = 10000       # Waste stream 1 (m^3/d)
Q2 = 15000       # Waste stream 2 (m^3/d)

DOr = 7.5        # DO concentration in river inflow (mg/L)
```

```
DO1 = 5          # DO concentration in waste stream 1 (mg/L)
DO2 = 5          # DO concentration in waste stream 2 (mg/L)

CBODr = 5        # CBOD concentration in river inflow (mg/L)
CBOD1 = 50       # CBOD concentration in waste stream 1 (mg/L)
CBOD2 = 45       # CBOD concentration in waste stream 2 (mg/L)

NBODr = 5        # NBOD concentration in river inflow (mg/L)
NBOD1 = 35       # NBOD concentration in waste stream 1 (mg/L)
NBOD2 = 35       # NBOD concentration in waste stream 2 (mg/L)

# Mixing at point 1
DO_0 = (Qr*DOr + Q1*DO1)  / (Qr+Q1)
CBOD_0 = (Qr*CBODr + Q1*CBOD1)  / (Qr+Q1)
NBOD_0 = (Qr*NBODr + Q1*NBOD1)  / (Qr+Q1)
Q_mix = Qr + Q1

function decay(x,U,C0,B0,N0)
    # x = distance d km downstream
    a1 = exp(-ka*x/U)
    a2 = ((kc/(ka-kc)) * (exp(-kc*x/U) - exp(-ka*x/U)))
    a3 = ((kn/(ka-kn)) * (exp(-kn*x/U) - exp(-ka*x/U)))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    return Cx                                              # DO concentration at distance x
end

# Values at 15 km before mixing
DO_first = decay(15, v, DO_0, CBOD_0, NBOD_0)
CBOD_first = CBOD_0 * exp(-kc * 15 / v)
NBOD_first  = NBOD_0  * exp(-kn * 15 / v)

# Mixing with wastewater stream 2
DO_1 = (Q_mix*DO_first + Q2*DO2)  / (Q_mix + Q2)
NBOD_1 = (Q_mix*NBOD_first + Q2*NBOD2)  / (Q_mix + Q2)
CBOD_1 = (Q_mix*CBOD_first + Q2*CBOD2)  / (Q_mix + Q2)

# Plot for 0 to 50 km
xs = 0:1:50
DO_vals = Float64[]

for i in xs
    if i <= 15
```

```julia
        push!(DO_vals, decay(i, v, DO_0, CBOD_0, NBOD_0))
    else
        push!(DO_vals, decay(i - 15, v, DO_1, CBOD_1, NBOD_1))
    end
end

# Identify and plot minimum DO
min_DO, index = findmin(DO_vals)
min_x = xs[index]

println("Analytic Solution: Minimum DO = $(round(min_DO, digits=2)) mg/L at $(min_x) km")

# plot the DO profile
plot(xs, DO_vals,
     xlabel="Distance downstream (km)",
     ylabel="DO (mg/L)",
     title="Dissolved Oxygen Along River",
     lw=2)

scatter!([min_x], [min_DO],
         color=:red,
         markersize=:6,
         label="Minimum DO")

# annotate!([(min_x, min_DO, text("Min DO = $(round(min_DO, digits=2)) mg/L at $(min_x) km",
```

```
Analytic Solution: Minimum DO = 3.76 mg/L at 22 km
```



Problem 1.2

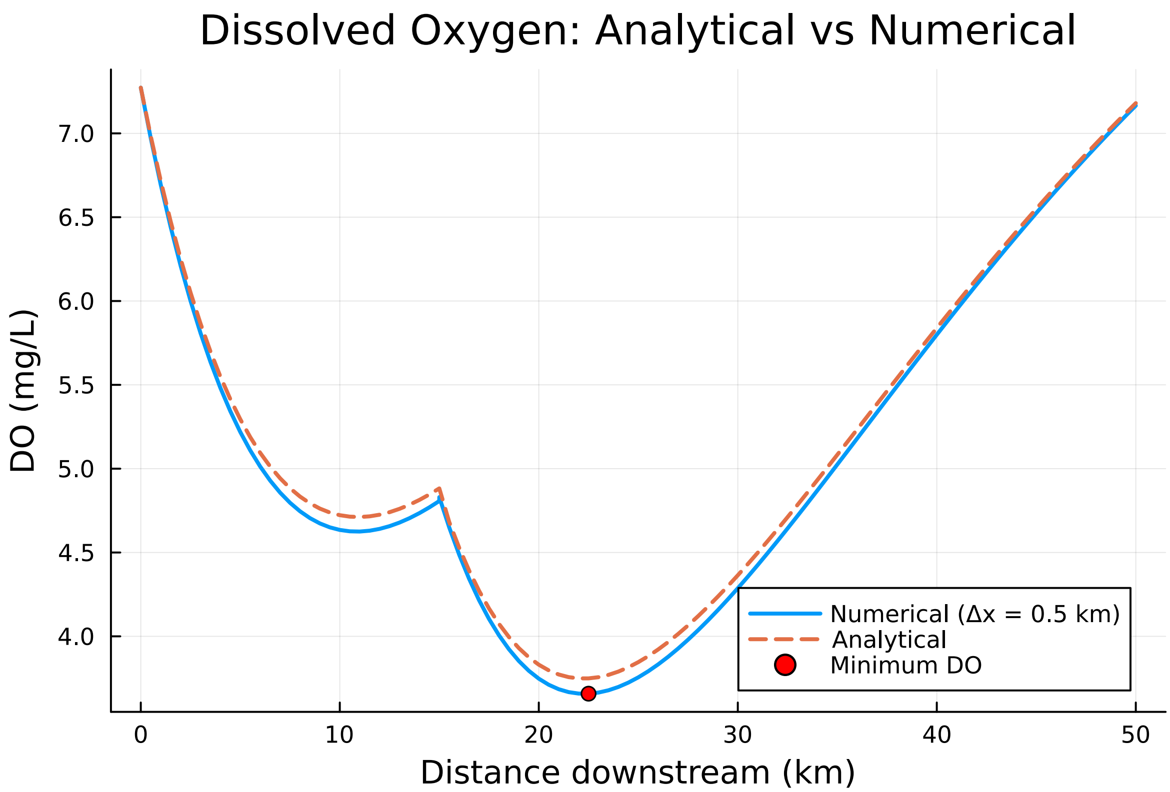
Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
function DO_simulate(DO0, CBOD0, NBOD0, v, ka, kc, kn, Cs, Δx, xmax)
    nsteps = Int(xmax / Δx)
    xs = collect(0:Δx:xmax)
    DO = zeros(length(xs))
    CBOD = zeros(length(xs))
    NBOD = zeros(length(xs))

    DO[1] = DO0
    CBOD[1] = CBOD0
    NBOD[1] = NBOD0
```

```
    for i in 1:(nsteps)
        dDOdx = (ka/v)*(Cs - DO[i]) - (kc/v)*CBOD[i] - (kn/v)*NBOD[i]
        DO[i+1] = DO[i] + Δx * dDOdx
        CBOD[i+1] = CBOD[i] * exp(-kc*Δx/v)
        NBOD[i+1] = NBOD[i] * exp(-kn*Δx/v)
    end
    return xs, DO, CBOD, NBOD
end

# segment 1: 0-15 km - first waste stream
xs1, DO1_vals, CBOD1_vals, NBOD1_vals = DO_simulate(DO_0, CBOD_0, NBOD_0, v, ka, kc, kn, Cs,

# mixing at 15 km with waste stream 2
DO_15 = DO1_vals[end]
CBOD_15 = CBOD1_vals[end]
NBOD_15 = NBOD1_vals[end]

DO_mix2 = (Q_mix*DO_15 + Q2*DO2) / (Q_mix + Q2)
CBOD_mix2 = (Q_mix*CBOD_15 + Q2*CBOD2) / (Q_mix + Q2)
NBOD_mix2 = (Q_mix*NBOD_15 + Q2*NBOD2) / (Q_mix + Q2)

#Segment 2 15-50 km
xs2, DO2_vals, CBOD2_vals, NBOD2_vals = DO_simulate(DO_mix2, CBOD_mix2, NBOD_mix2, v, ka, kc

# Combine both reaches
xs_total = vcat(xs1, xs2 .+ 15)
DO_total = vcat(DO1_vals, DO2_vals)

xs_analytic = 0:0.5:50
DO_analytic = [x <= 15 ? decay(x, v, DO_0, CBOD_0, NBOD_0) :
                          decay(x - 15, v, DO_1, CBOD_1, NBOD_1) for x in xs_analytic]

xs_num = xs_total
DO_num = DO_total

#Plot both solutions
p = plot(xs_num, DO_num, lw=2, label="Numerical (Δx = 0.5 km)",
         xlabel="Distance downstream (km)", ylabel="DO (mg/L)",
         title="Dissolved Oxygen: Analytical vs Numerical")

plot!(p, xs_analytic, DO_analytic, lw=2, linestyle=:dash, label="Analytical")
```

```julia
# Identify and plot minimum DO in numerical solution
min_DO, idx = findmin(DO_num)
println("Minimum DO = $(round(min_DO, digits=3)) mg/L at x = $(xs_num[idx]) km")
scatter!(p, [xs_num[idx]], [min_DO], color=:red, label="Minimum DO")

display(p)
```

```
Minimum DO = 3.658 mg/L at x = 22.5 km
```



The minimum value of the numerical solution was less than the analytic solution by .102mg/L. The difference is caused by truncation error, which arises from using numerical methods to simply an analytical solution. The numerical solution was found through a first-order Taylor expansion, neglecting other order terms.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a $\text{LogNormal}(2.0, 0.15)$ distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.